

ELDOR CORPORATION

FREIGHT ROUTE EVALUATION SURVEY

CONSOLIDATED ANALYSIS REPORT

All 32 Respondents · All 3 Data Waves · Final Dataset

32

Total Respondents

461

Route Evaluations

7

Trade Lanes

18

Evaluation Criteria

3

Data Collection Waves

8,200

Total Scores

Diploma Projesi: **2025 / 2026**

Academic Advisor: **Prof. Dr. Gül Denктаş Şakar**

Trade Lanes: **China Import · Italy Export · Italy Import · Germany Export**

Score Scale: **1 = Very Poor / Very High Risk → 5 = Excellent / Very Low Risk**

Statistical Tests: **t-tests · Pearson correlation · Descriptive statistics**

Report Date: **16 May 2026**

Wave 1 (N=10) · Wave 2 (N=15) · Wave 3 (N=7)

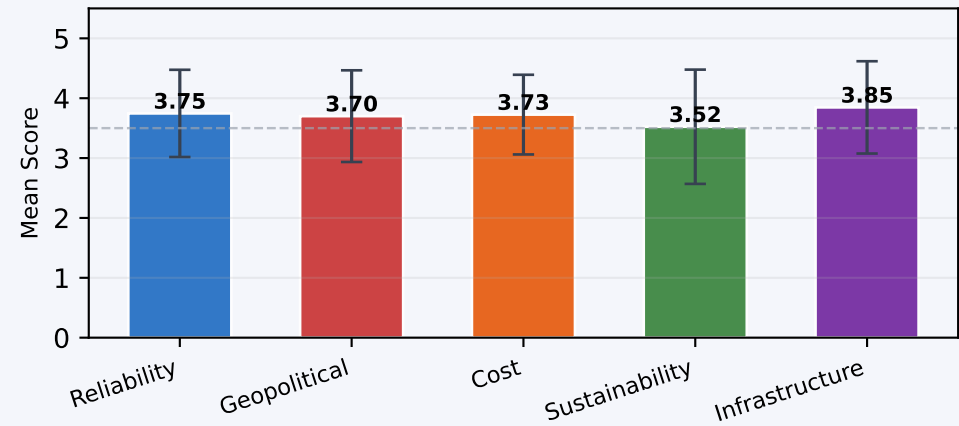
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EXECUTIVE SUMMARY

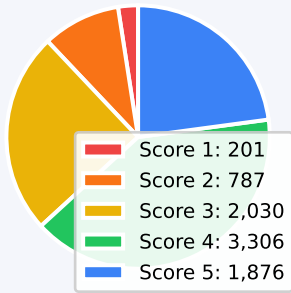
Complete dataset: 31 named respondents + Eldor_FIXED (duplicate EN file) | 461 evaluations | 8,200 individual scores

TradeLane	N	Mean	Std	Median	Min	Max	Rating
Italy Export Main	31	4.05	0.34	4.06	3.22	4.72	EXCELLENT
Italy Import Main	31	4.01	0.32	4.0	3.18	4.83	EXCELLENT
Italy Export Alt	93	3.83	0.5	3.83	2.72	4.94	GOOD
Germany Export Alt	93	3.81	0.55	3.89	1.94	5.0	GOOD
Italy Import Alt	62	3.8	0.58	3.83	2.0	4.89	GOOD
Germany Export Main	31	3.57	0.67	3.83	2.17	4.89	GOOD
China Import	120	3.38	0.6	3.39	1.83	4.89	MODERATE

Category Performance Overview



Likert Score Distribution
(Total: 8,200 individual scores)

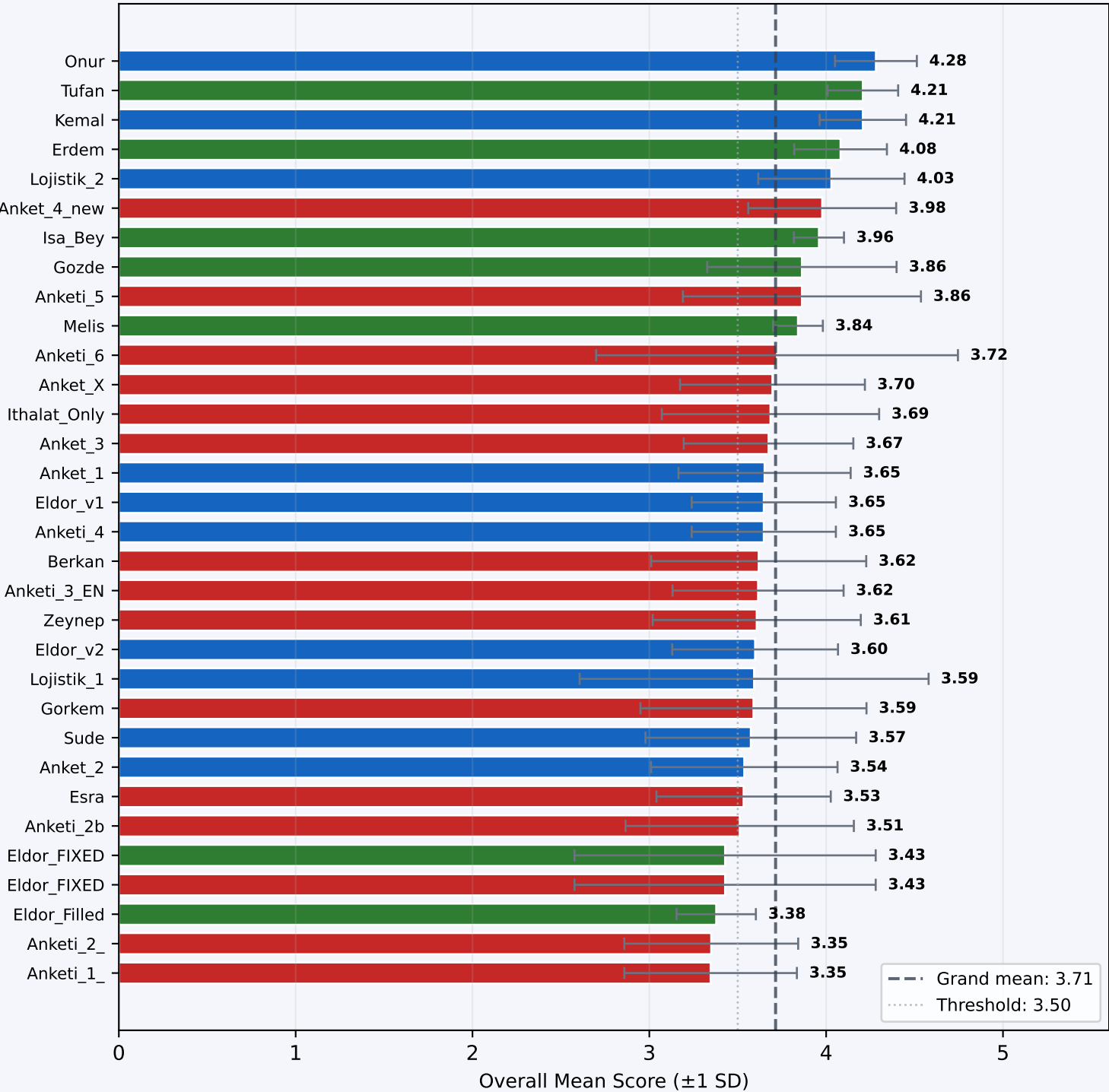


KEY FINDINGS

- Italy Export Main is the top-ranked trade lane (4.05/5.0) — confirmed across all 3 waves with lowest variability (SD=0.34).
- Suez Canal Route is critically weak on geopolitical safety: mean 2.50 vs Cape mean 3.70— gap is highly significant ($t=-7.09$, $p<0.001$, $n=89$).
- Green Options is the weakest criterion across the entire dataset (mean 3.35)— persistent sustainability gap.
- Italy Export Main significantly outperforms Germany Export Main ($t=3.49$, $p=0.0009$, ***)— Italy corridor is statistically superior.
- Rail Route (Trans-Caspian/BRI) has lowest overall score (3.26/5) despite best sustainability— infrastructure & risk remain blockers.
- Score 4 ("Good") is the modal response: 40.3% of all 8,200 scores— respondents lean moderately positive across the dataset.

ALL RESPONDENTS — Individual Score Overview (N=32)

Each bar = respondent overall mean across all trade lanes and routes they evaluated



Data Waves



Wave 1 (N=10)

Sude, Anket_1/2, Kemal, Lojistik_1/2,
Anketi_4, Eldor_v1/v2, Onur



Summary Statistics

Wave 2 (N=15)

Grand Mean **3.714**

Anket_3/4/X, Anketi_1_/2_/2b/3EN/5/6,

Berkan, Esra, Berkan, Zeynep, İthalat, **0.586**

Eldor_FIXED (x2)

Min Score **1.833**

Max Score **5.000**

Median Score **3.833**

Q1 (25%) **3.389**

Q3 (75%) **Wave 3 (N=7)** **4.062**

Erdem, Gözde, İsa Bey, Melis, Tufan,

Total Evaluations **461**

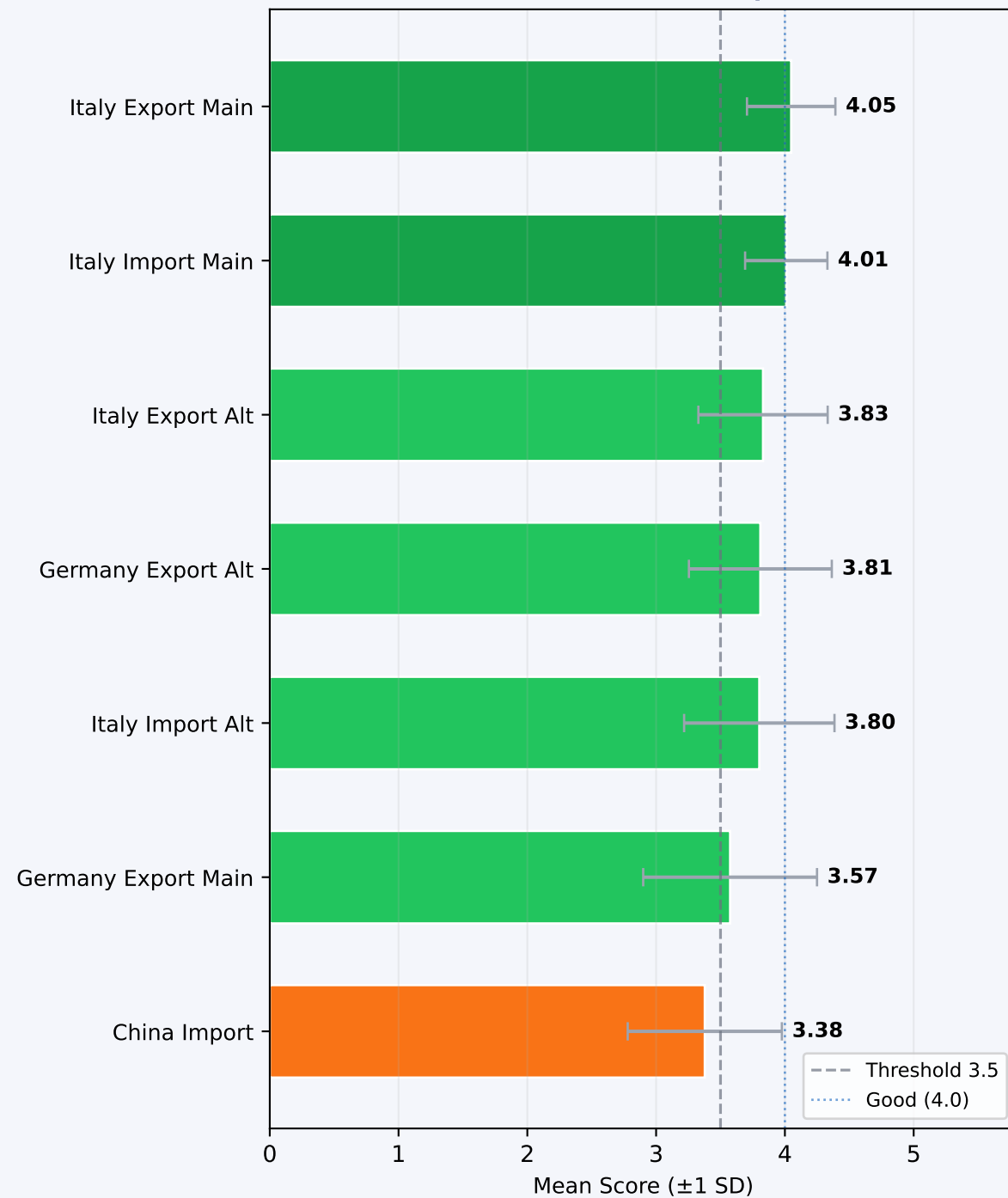
Eldor_FIXED, Eldor_Filled

Total Criteria Scores **8,200**

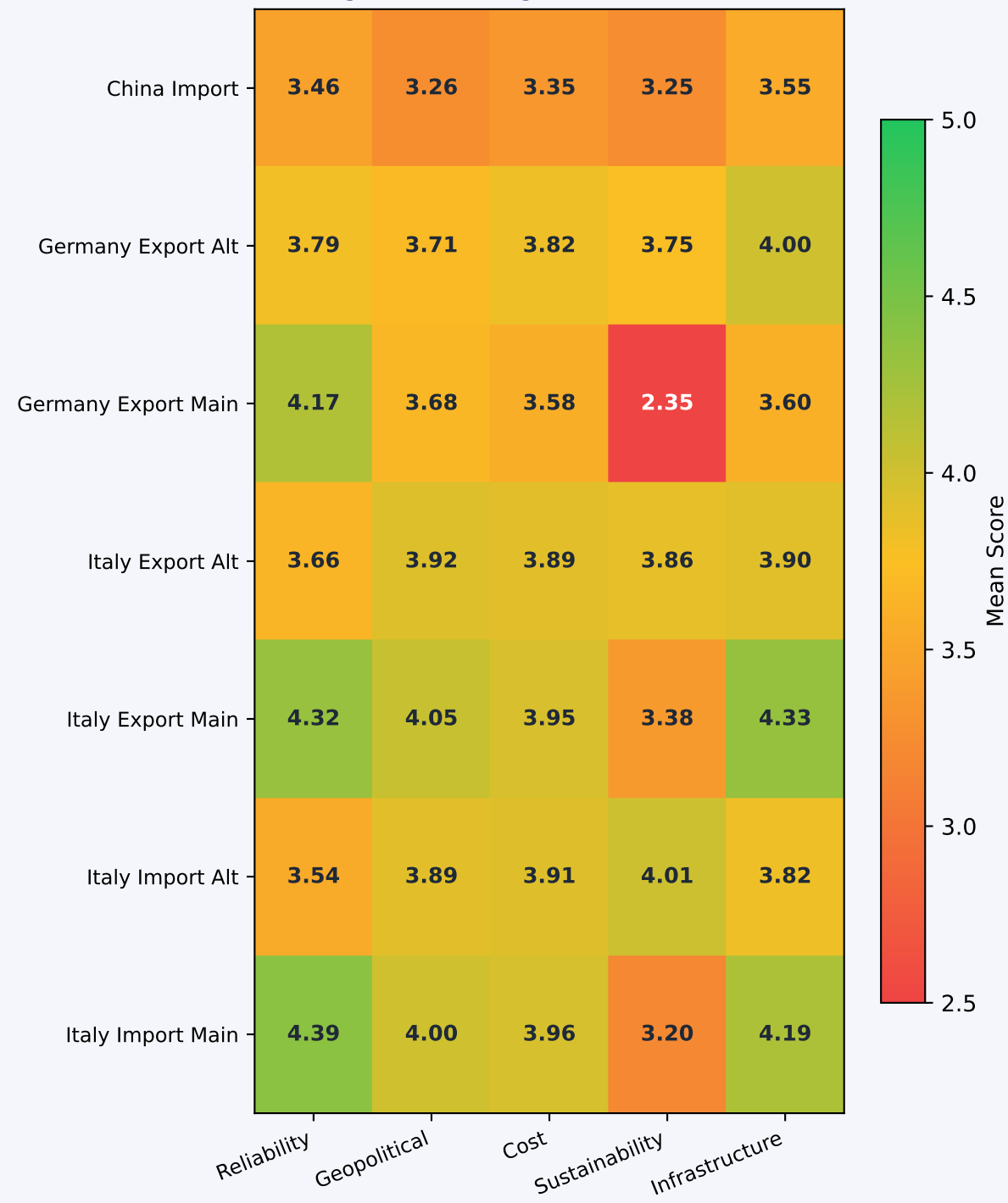
Score range **1.83 - 5.00**

TRADE LANE ANALYSIS — Rankings & Category Heatmap

Trade Lane Rankings
(all 461 evaluations, 32 respondents)

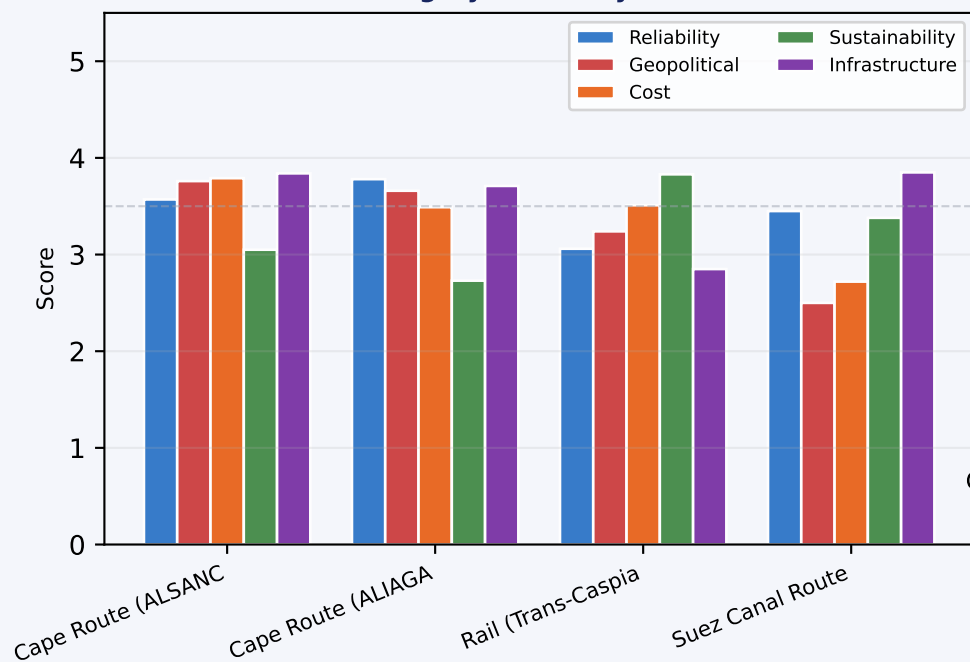


Category Score Heatmap by Trade Lane
(green = stronger · red = weaker)

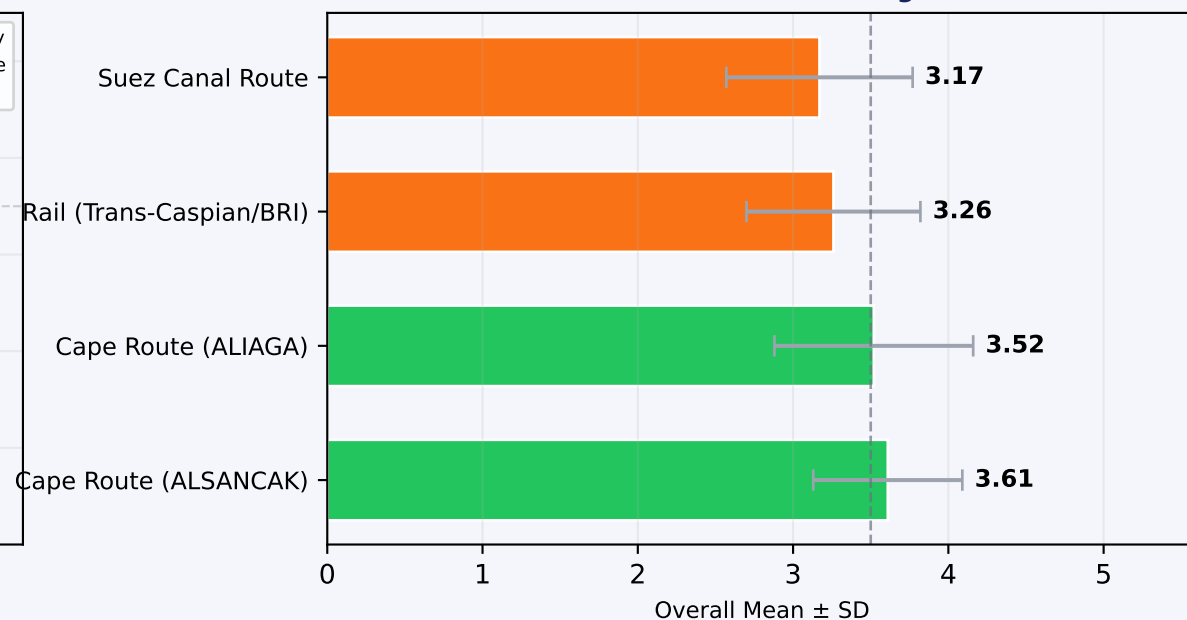


CHINA IMPORT — Complete Route Evaluation (N=32 respondents, 120 evaluations)

Category Scores by Route

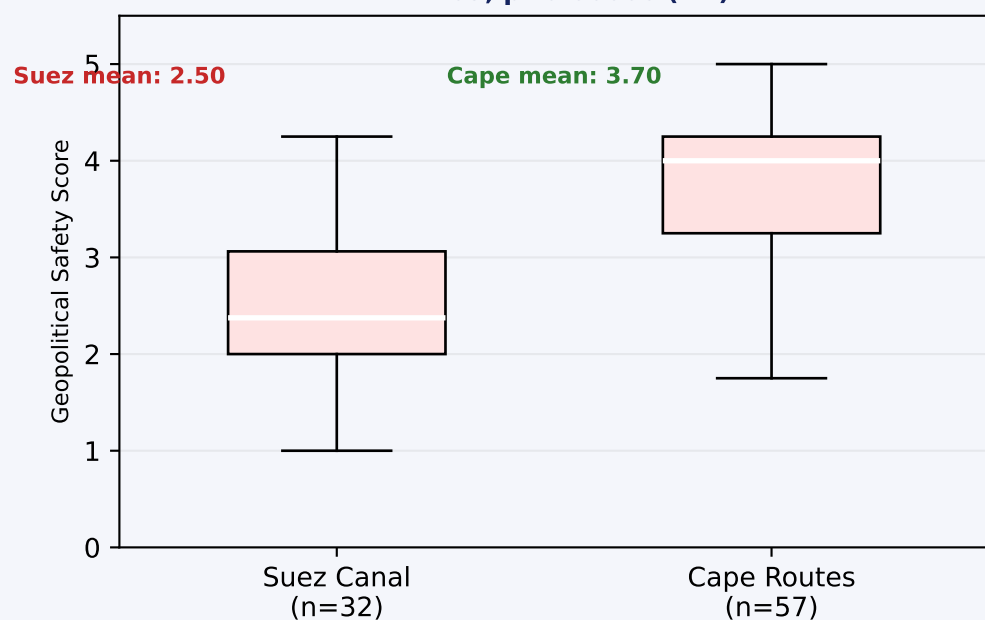


Overall Route Rankings



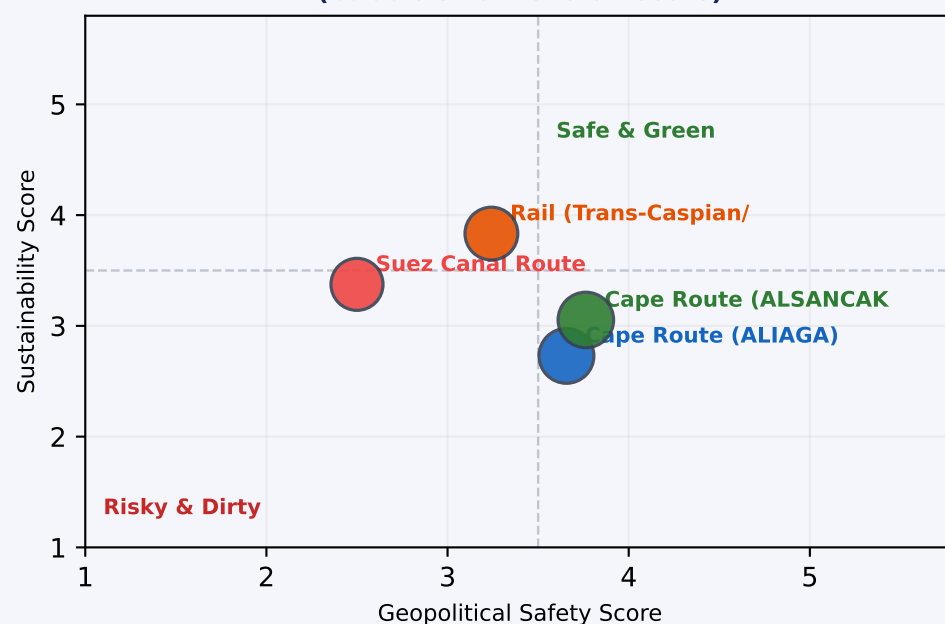
Suez vs Cape: Geopolitical Safety

t=-7.09, p=0.00000 (***)



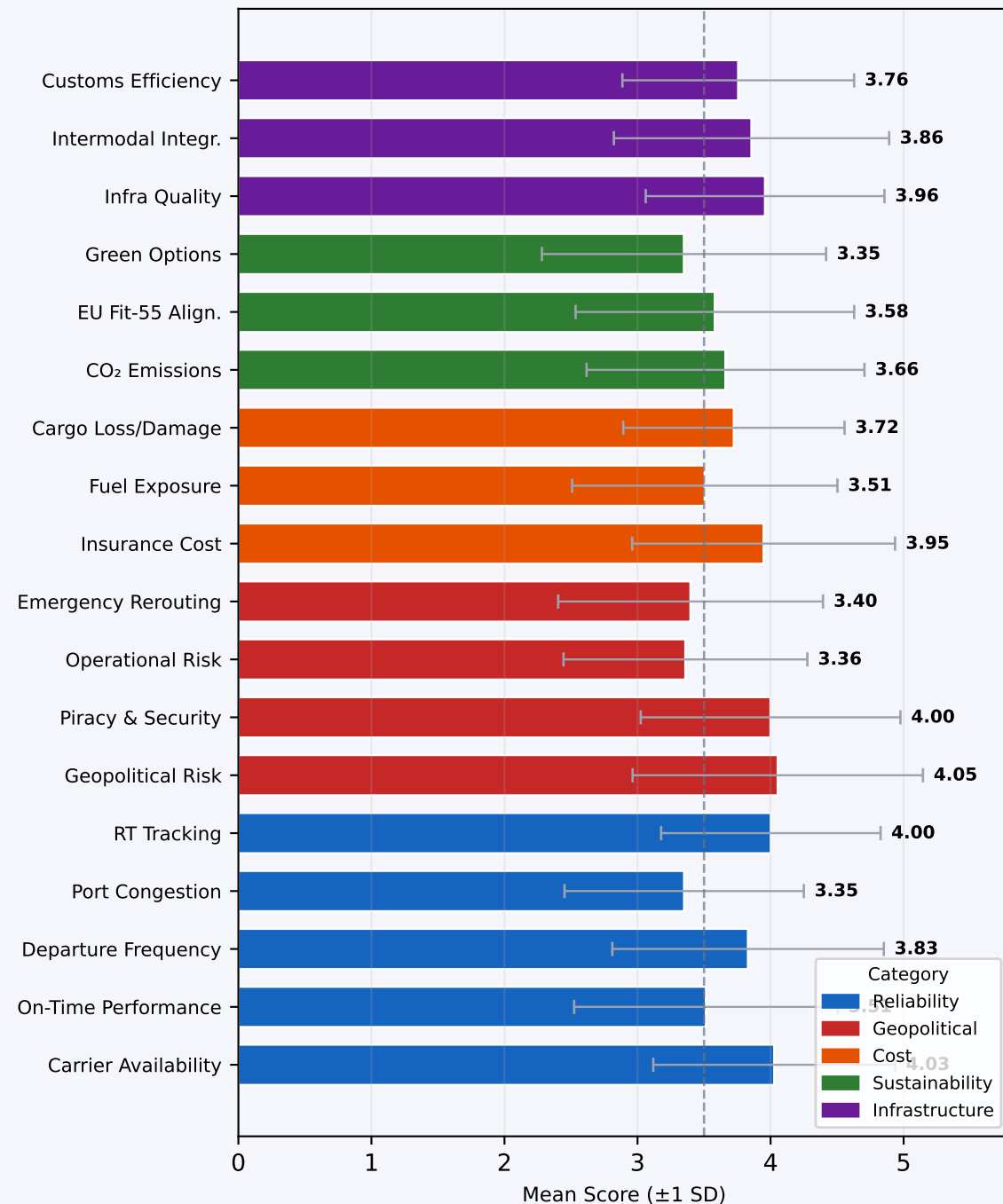
Strategic Quadrant: Safety vs Sustainability

(bubble size ∝ overall score)

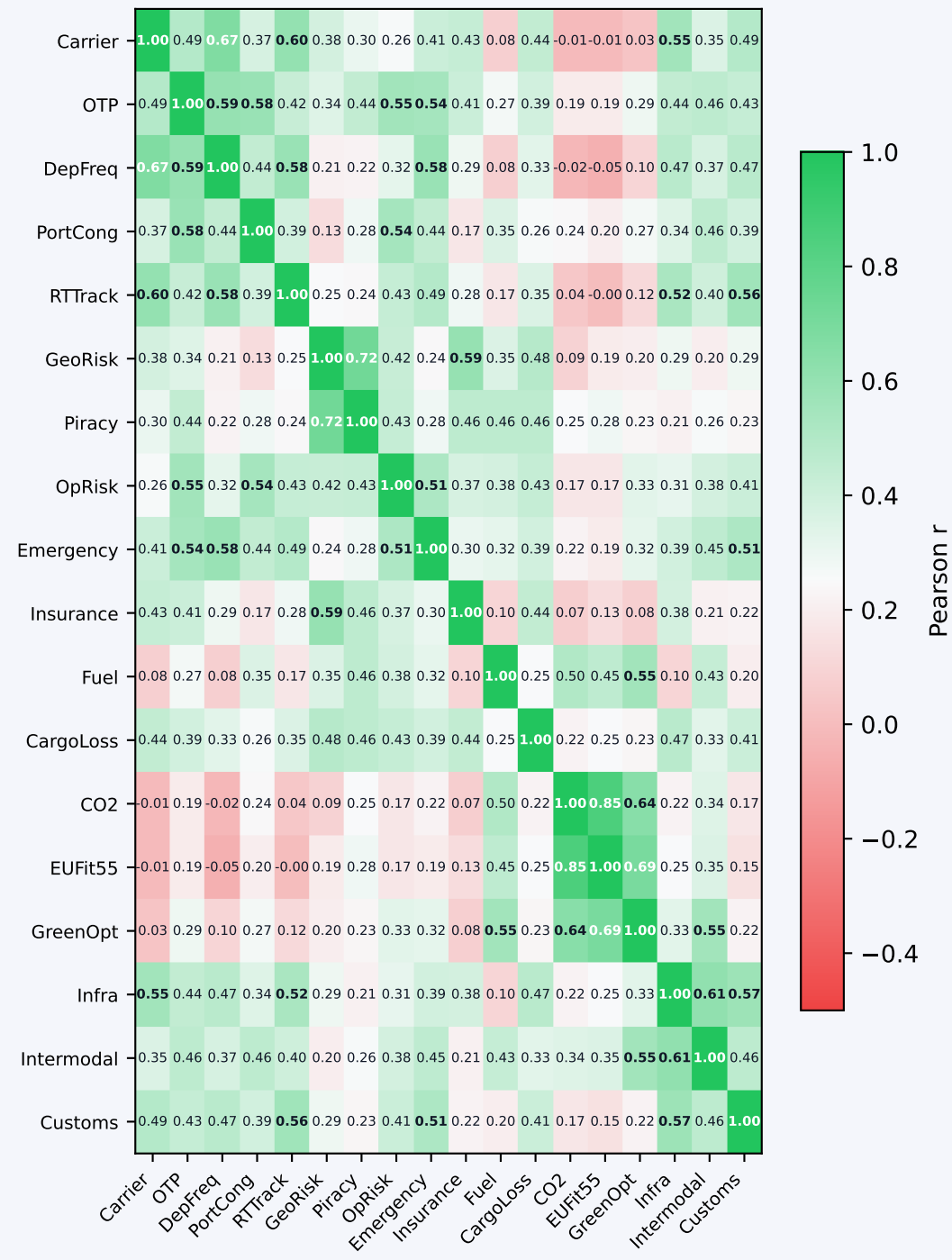


CRITERIA ANALYSIS — All 18 Dimensions (N=461 evaluations)

All Criteria — Mean ± 1 SD
(colour = category)

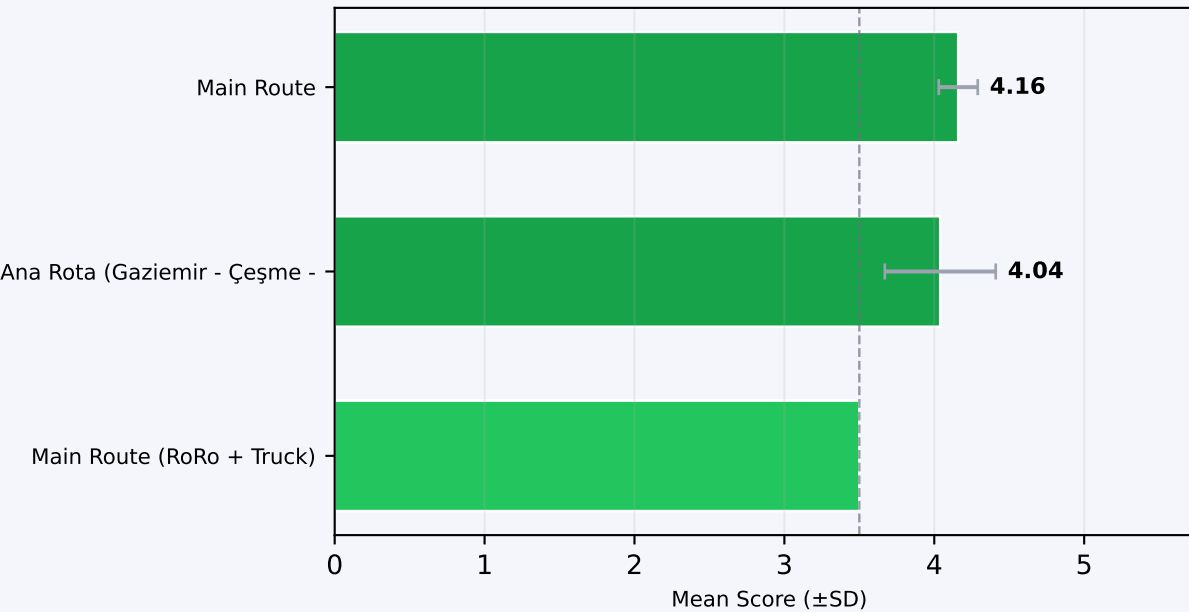


Criteria Correlation Heatmap
(N=461 | Green=positive, Red=negative)

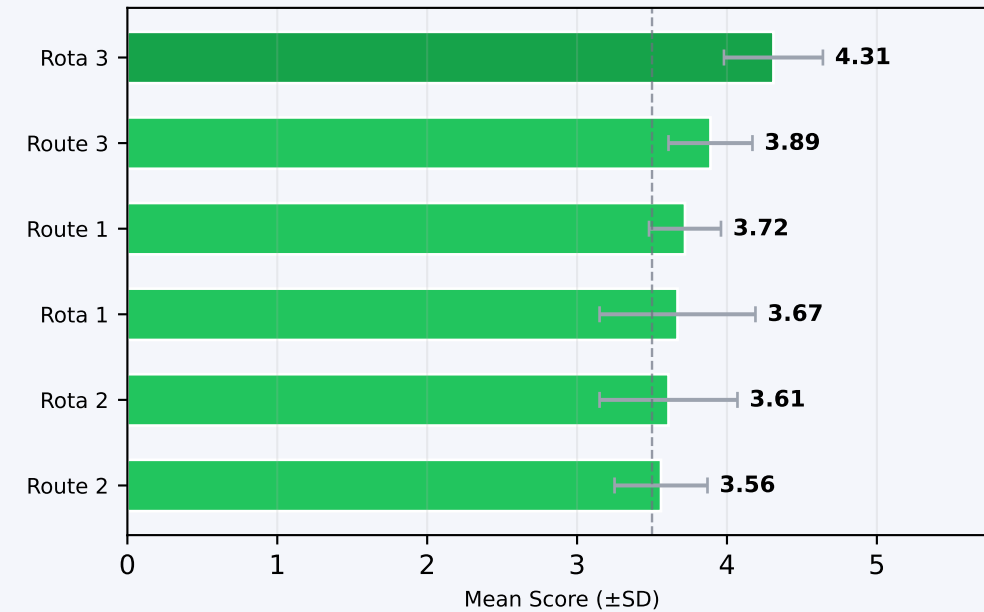


EXPORT & IMPORT ROUTE COMPARISONS — Italy & Germany

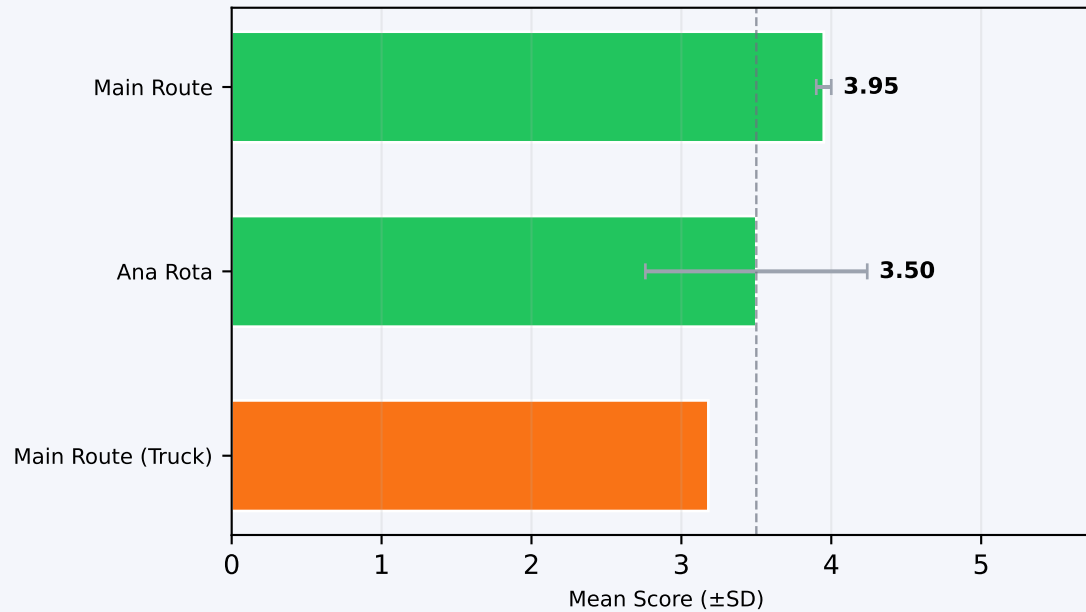
Italy Export — Main Route (N=31 evaluations)



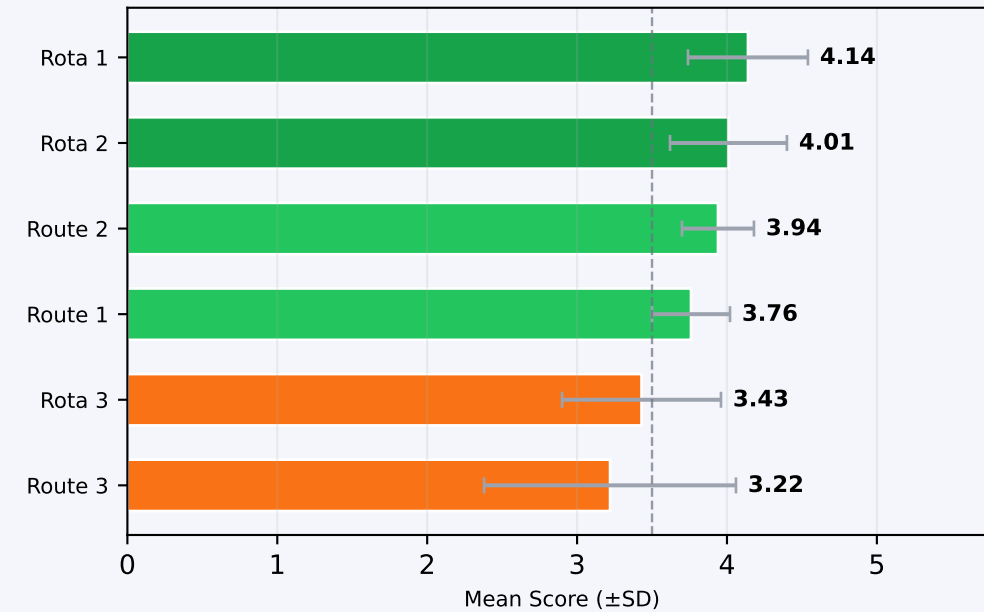
Italy Export — Alternative Routes (N=93 evaluations)



Germany Export — Main Route (N=31 evaluations)

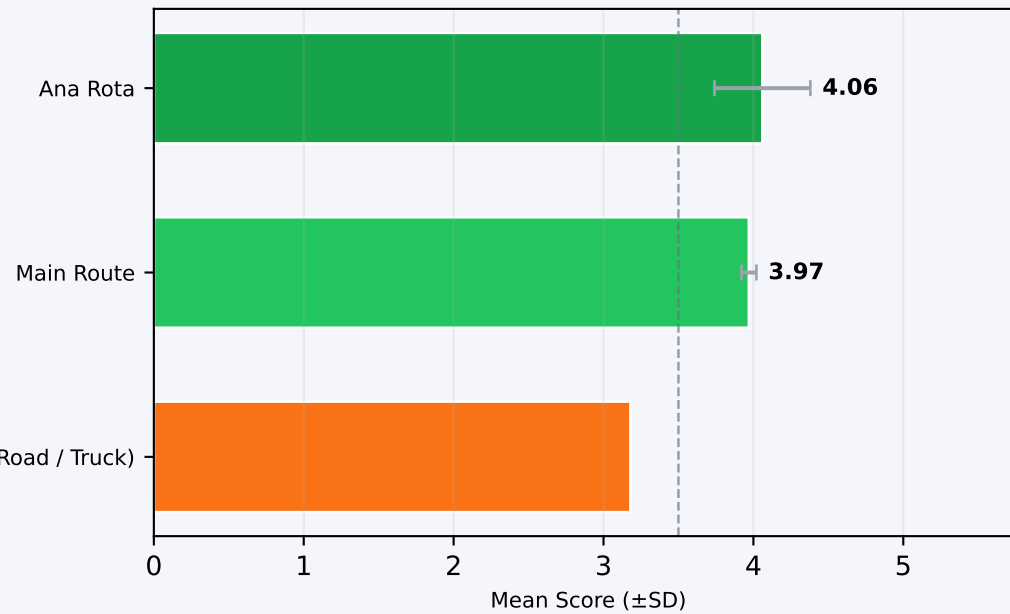


Germany Export — Alternative Routes (N=93 evaluations)

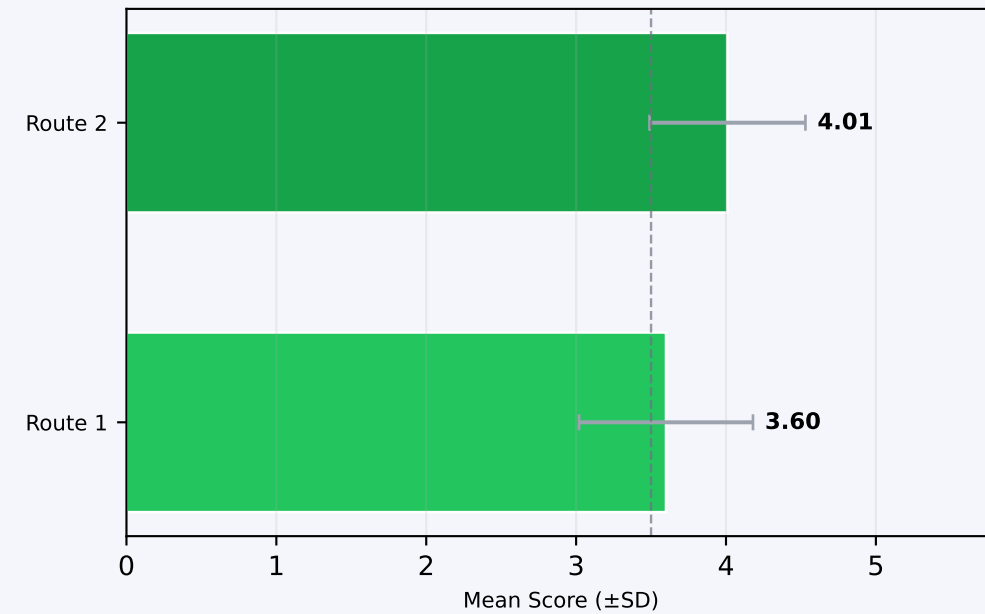


ITALY IMPORT ROUTES + STATISTICAL TESTS

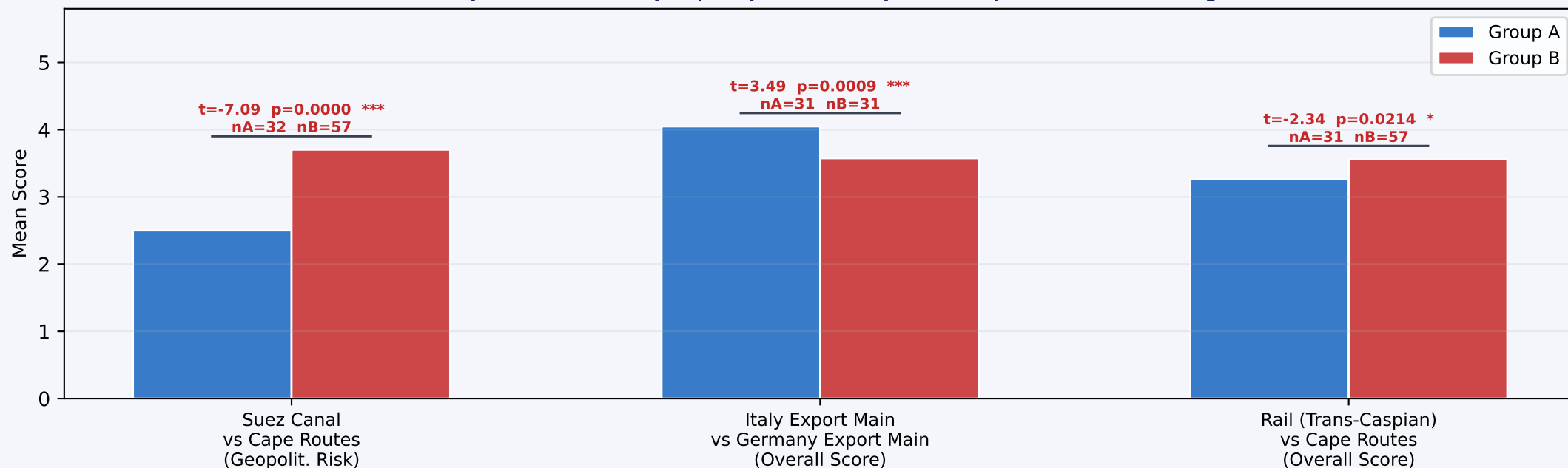
Italy Import — Main Route
(N=31 evaluations)



Italy Import — Alternative Routes
(N=62 evaluations)

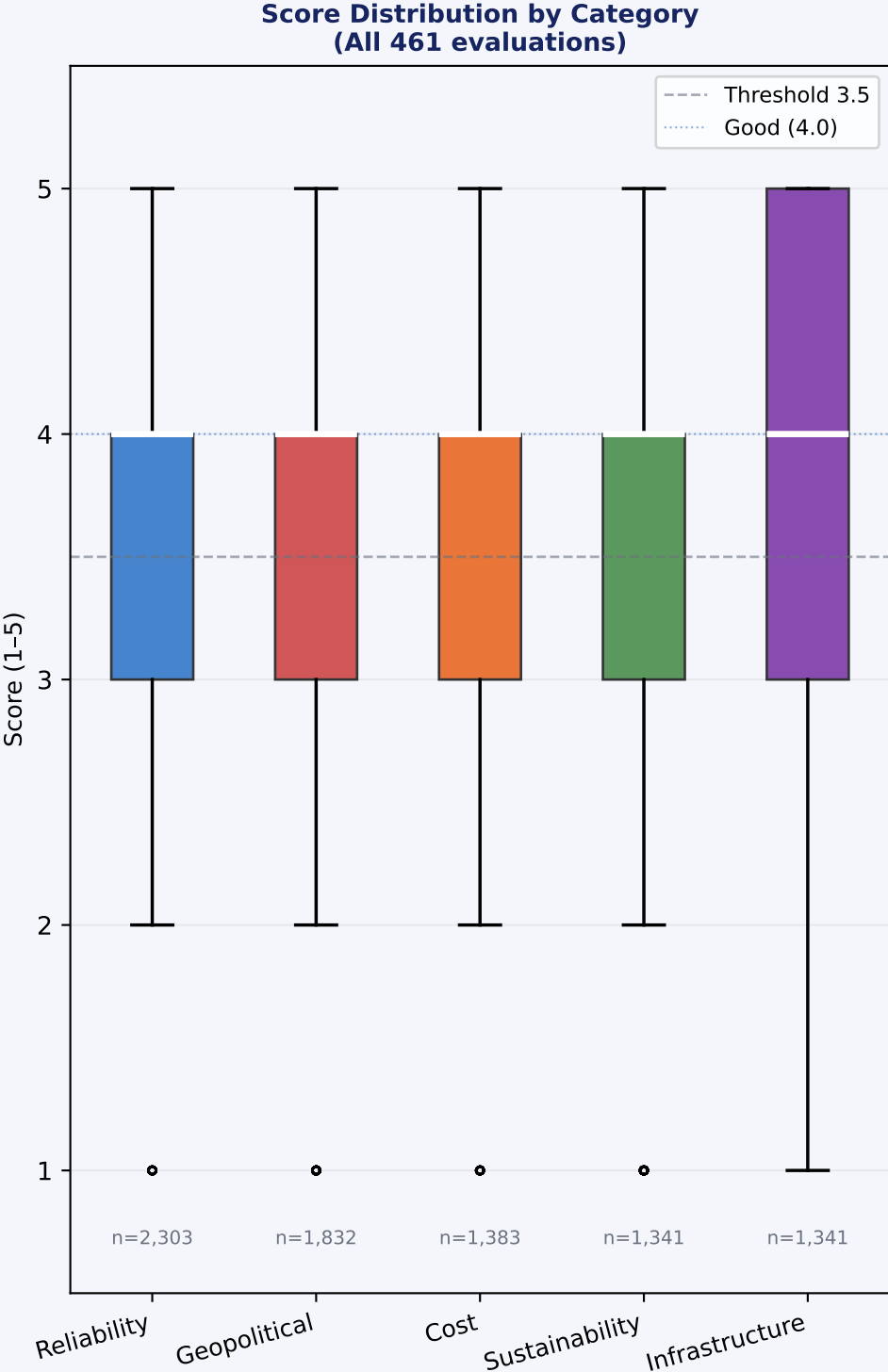


Statistical Tests — t-Test Results
(Blue = Group A / Red = Group B | *** p<0.001 ** p<0.01 * p<0.05 ns = not significant)



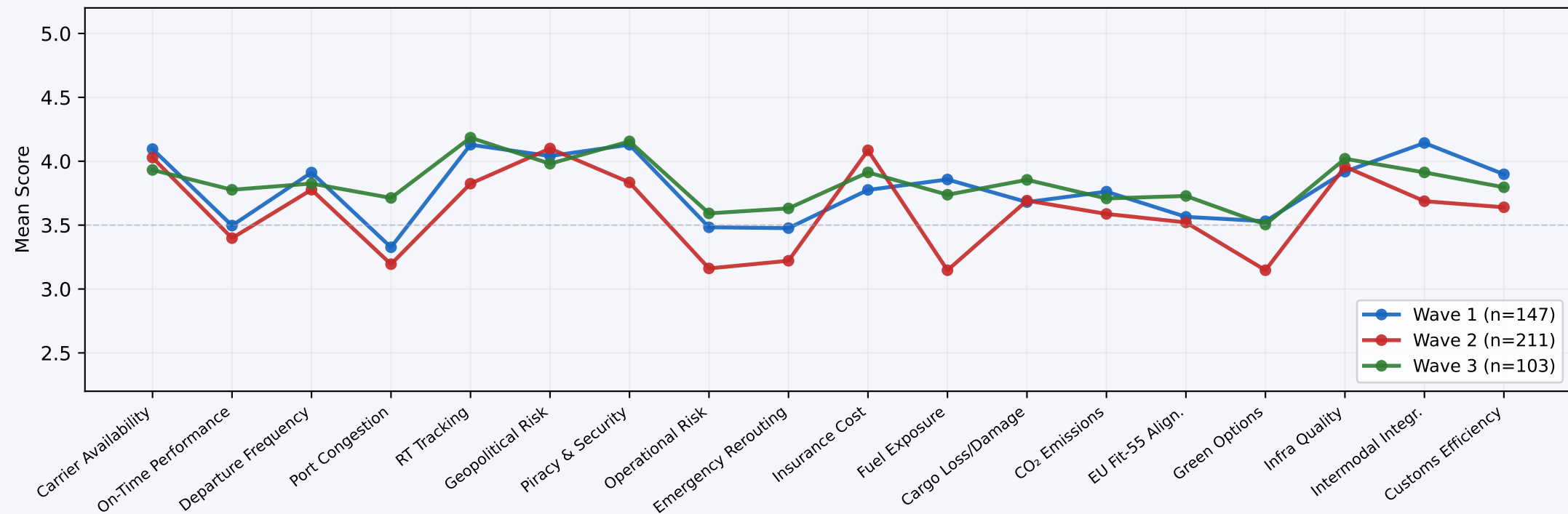
DESCRIPTIVE STATISTICS — All 18 Criteria + Category Distributions

Criterion	Mean	Std	Min	Q1	Med	Q3	Max
Availability	4.03	0.91	1.0	4.0	4.0	5.0	5.0
Performance	3.51	0.99	1.0	3.0	4.0	4.0	5.0
Frequency	3.83	1.02	1.0	3.0	4.0	5.0	5.0
Congestion	3.35	0.9	1.0	3.0	3.0	4.0	5.0
RT Tracking	4.0	0.82	2.0	4.0	4.0	5.0	5.0
Political Risk	4.05	1.09	1.0	4.0	4.0	5.0	5.0
Cyber Security	4.0	0.98	1.0	4.0	4.0	5.0	5.0
Operational Risk	3.36	0.92	1.0	3.0	3.0	4.0	5.0
Dynamic Rerouting	3.4	1.0	1.0	3.0	3.0	4.0	5.0
Insurance Cost	3.95	0.99	1.0	3.0	4.0	5.0	5.0
Climate Exposure	3.51	1.0	1.0	3.0	4.0	4.0	5.0
Loss/Damage	3.72	0.83	1.0	3.0	4.0	4.0	5.0
Emissions	3.66	1.04	1.0	3.0	4.0	4.0	5.0
Net-55 Align.	3.58	1.05	1.0	3.0	4.0	4.0	5.0
Open Options	3.35	1.07	1.0	3.0	3.0	4.0	5.0
Infra Quality	3.96	0.9	1.0	3.0	4.0	5.0	5.0
Modal Integr.	3.86	1.03	1.0	3.0	4.0	5.0	5.0
Logistics Efficiency	3.76	0.87	1.0	3.0	4.0	4.0	5.0

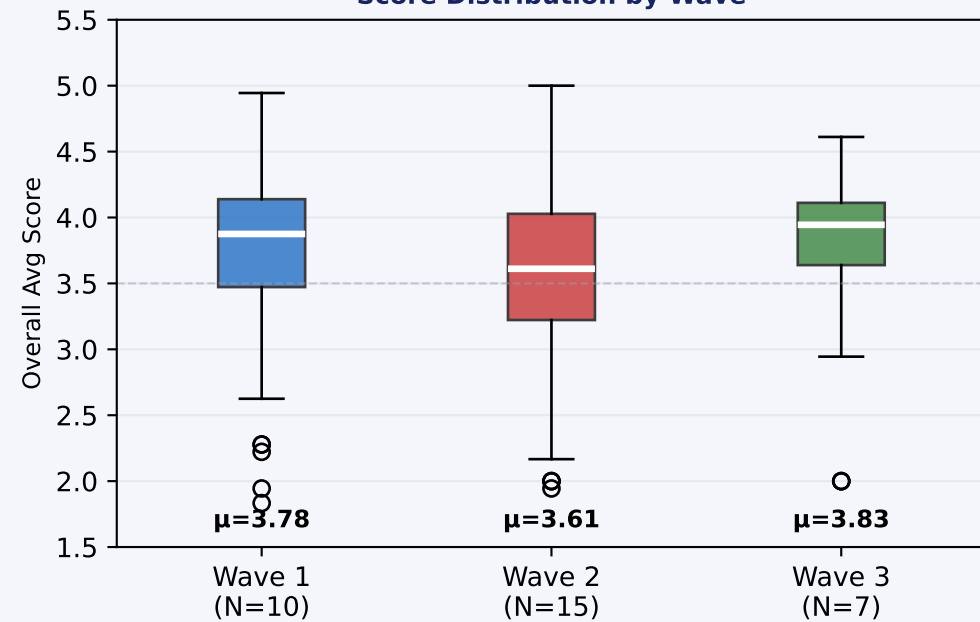


LONGITUDINAL TRENDS — Three Waves + Likert Distribution

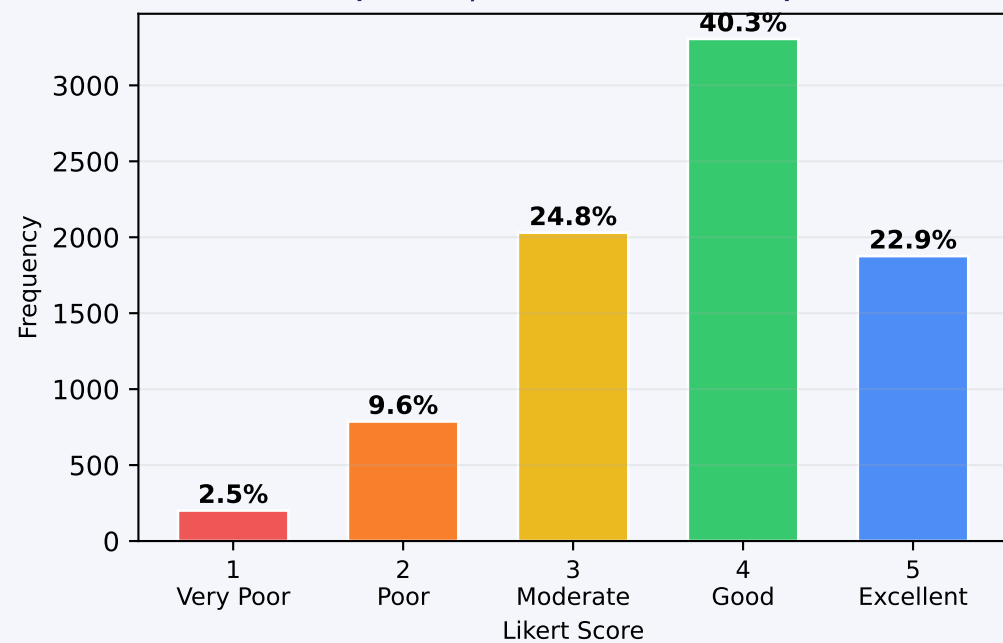
Criteria Score Evolution Across Three Waves



Score Distribution by Wave



Likert Score Frequency Distribution (Total: 8,200 individual scores)



FINAL RECOMMENDATIONS & CONCLUSIONS

Based on 32 respondents · 3 data waves · 461 route evaluations · 8,200 individual Likert scores

Trade Lane	Recommended Route	Score	Rating
Italy Export Alt	Rota 3	4.31	EXCELLENT ★★★
Italy Export Main	Main Route	4.16	EXCELLENT ★★★
Germany Export Alt	Rota 1	4.14	EXCELLENT ★★★
Italy Import Main	Ana Rota	4.06	EXCELLENT ★★★
Italy Import Alt	Route 2	4.01	EXCELLENT ★★★
Germany Export Main	Main Route	3.95	GOOD ★★
China Import	Konteyner Gemi Rotası 02 - ALIAGA	3.7	GOOD ★★

ROUTE STRATEGY

- USE Italy Export Main (RoRo+Truck, Gaziemir-Trieste) as primary export lane.
- USE Cape/ALIAGA or Cape/ALSANCAK for China imports (both score 3.35/5).
- AVOID Suez Canal for China: geopolitical score 2.50 vs Cape 3.70 ($p<0.0001$).
- CONSIDER Italy Export Alt Route 3 (RoRo+Rail+Truck) — top alt at 4.31.
- USE Germany Export Alt Route 1 (RoRo+Rail+Truck) for Germany — 4.14.

RISK PRIORITIES

- MITIGATE Port Congestion risk (3.35/5) — consistently weakest in Reliability.
- ADDRESS Green Options gap (3.35/5) — lowest criterion in entire dataset.
- REDUCE Operational Risk (3.36) and Emergency Rerouting (3.40) weaknesses.
- AVOID Rail as primary lane: infra score 2.85, lowest of all China routes.
- TRACK OTP Performance (3.51) — 2nd weakest criterion, supply-chain post-2024.

THESIS ARGUMENTS

- Suez vs Cape: $t=-7.09$, $p<0.001$, $n=89$ — strongest finding in Reliability.
- Italy vs Germany Export Main: $t=3.49$, $p=0.0009$ — statistically significant.
- 3-wave design confirms finding stability — strengthens methodological validity.
- 8,200 individual scores from 32 experts provide robust quantitative base.
- Safety-Sustainability quadrant (Ch.Import) directly supports post-2024 thesis.